Foundations of Data Science Project - Diabetes Analysis

Context

Diabetes is one of the most frequent diseases worldwide and the number of diabetic patients are growing over the years. The main cause of diabetes remains unknown, yet scientists believe that both genetic factors and environmental lifestyle play a major role in diabetes.

A few years ago research was done on a tribe in America which is called the Pima tribe (also known as the Pima Indians). In this tribe, it was found that the ladies are prone to diabetes very early. Several constraints were placed on the selection of these instances from a larger database. In particular, all patients were females at least 21 years old of Pima Indian heritage.

Objective

Here, we are analyzing different aspects of Diabetes in the Pima Indians tribe by doing Exploratory Data Analysis.

Data Dictionary

The dataset has the following information:

- Pregnancies: Number of times pregnant
- Glucose: Plasma glucose concentration over 2 hours in an oral glucose tolerance test
- BloodPressure: Diastolic blood pressure (mm Hg)
- SkinThickness: Triceps skin fold thickness (mm)
- Insulin: 2-Hour serum insulin (mu U/ml)
- BMI: Body mass index (weight in kg/(height in m)^2)
- DiabetesPedigreeFunction: A function that scores the likelihood of diabetes based on family history.
- Age: Age in years
- Outcome: Class variable (0: a person is not diabetic or 1: a person is diabetic)

Q 1: Import the necessary libraries and briefly explain the use of each library (3 Marks)

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In [1]:
```
# remove _____ & write the appropriate library name

import numpy as np
import pandas as pd

import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

Write your Answer here:

Ans 1: import numpy as np import pandas as pd import seaborn as sns import matplotlib.pyplot as plt # np is one of the ways to abbreviate numpy, a library that helps us to use mathematical functions # pandas is a library very useful to do exploratory analysis, the type of analysis we are going to do for the pima indians diabetes # matplotlib is the library that allows us to have high quality graphics. Very helpful for the visualization of certain variable # seaborn is the library that we need to use if we are going to do satistical data visualizations. This is fundamental to identify patterns and understand the behaviour of the variables

Q 2: Read the given dataset (1 Mark)

In [2]:
```
# remove _____ & write the appropriate function name

pima = pd.read_csv(r'C:\Users\andre\Downloads\diabetes.csv')
```

Q3. Show the last 10 records of the dataset. How many columns are there? (1 Mark)

In [3]:
```
# remove ______ and write the appropriate number in the function

pima.tail(10)
```

Out[3]:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction
758	1	106	76	20	79	37.5	0.197
759	6	190	92	20	79	35.5	0.278
760	2	88	58	26	16	28.4	0.766
761	9	170	74	31	79	44.0	0.403
762	9	89	62	20	79	22.5	0.142
763	10	101	76	48	180	32.9	0.171
764	2	122	70	27	79	36.8	0.340
765	5	121	72	23	112	26.2	0.245
766	1	126	60	20	79	30.1	0.349
767	1	93	70	31	79	30.4	0.315

Write your Answer here:

Ans 3: pima.tail(10) # Using tail to know the last 10 records. I found 9 columns.

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Q4. Show the first 10 records of the dataset (1 Mark)

In [4]:
```
# remove _____ & write the appropriate function name and the number of rows to get

pima.head(10) # Using head to know the first 10 records.
```

Out[4]:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunctic
0	6	148	72	35	79	33.600000	0.62
1	1	85	66	29	79	26.600000	0.35
2	8	183	64	20	79	23.300000	0.67
3	1	89	66	23	94	28.100000	0.16
4	0	137	40	35	168	43.100000	2.28
5	5	116	74	20	79	25.600000	0.20
6	3	78	50	32	88	31.000000	0.24
7	10	115	69	20	79	35.300000	0.13
8	2	197	70	45	543	30.500000	0.15
9	8	125	96	20	79	31.992578	0.23

Q5. What do you understand by the dimension of the dataset? Find the dimension of the `pima` dataframe. (1 Mark)

In [5]:
```
# remove _____ & write the appropriate function name

pima.shape
```

Out[5]:
```
(768, 9)
```

Write your Answer here:

Ans 5: pima.shape # The function shape helps us to know the dimension, which is the number of columns and rows of the dataframe. # (768, 9)

Q6. What do you understand by the size of the dataset? Find the size of the `pima` dataframe. (1 Mark)

In [6]:
```
# remove _____ & write the appropriate function name

pima.size
```

Out[6]:
```
6912
```

Write your Answer here:

Ans 6: pima.size # The function size helps us to know the number of elements along a given axis. # 6912

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Q7. What are the data types of all the variables in the data set? (2 Marks)

Hint: Use the info() function to get all the information about the dataset.

In [7]:
```
# remove _____ & write the appropriate function name

pima.info(verbose=True)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
```

Write your Answer here:

Ans 7: pima.info(verbose=True) # The data types are integer and float. # Seven variables are integer and two are float.

Q8. What do we mean by missing values? Are there any missing values in the `pima` dataframe? (2 Marks)

In [8]:
```
# remove _____ & write the appropriate function name

pima.isnull().values.any()
```

Out[8]:
```
False
```

Write your Answer here:

Ans 8: pima.isnull().values.any() # There are not any missing values, the answer is False.

Q9. What do the summary statistics of the data represent? Find the summary statistics for all variables except 'Outcome' in the `pima` data. Take one column/variable from the output table and explain all its statistical measures. (3 Marks)

In [9]:
```
# remove _____ & write the appropriate function name

pima.iloc[:,0:8].describe()
```

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Out[9]:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPe
count	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	
mean	3.845052	121.675781	72.250000	26.447917	118.270833	32.450805	
std	3.369578	30.436252	12.117203	9.733872	93.243829	6.875374	
min	0.000000	44.000000	24.000000	7.000000	14.000000	18.200000	
25%	1.000000	99.750000	64.000000	20.000000	79.000000	27.500000	
50%	3.000000	117.000000	72.000000	23.000000	79.000000	32.000000	
75%	6.000000	140.250000	80.000000	32.000000	127.250000	36.600000	
max	17.000000	199.000000	122.000000	99.000000	846.000000	67.100000	

Write your Answer here:

Ans 9: pima.iloc[:,0:8].describe() # The summary statistics is representing the number of elements (count), the mean/average (mean), the standard deviation (std), the minimum value (min), the 25th percentile (25%), the 50th percentile (50%), the 75th percentile (75%) and the maximum value (max) # In the variable age, we found that the average is approximately 33 years old. # The standard deviation is 11.76, so in the data set we will find people around 21.24 and 44.76. # That is in line with the minimum value of 21 years old. # The 25th percentile is the value at which 25% of the answers lie below that value, in this case, 24 years old. # The 50th percentile is the value at which 50% of the answers lie below that value, in this case, 29 years old. # The 75th percentile is the value at which 75% of the answers lie below that value, in this case, 41 years old. # We can infere that most of people in the dataframe are relatively young, they have less than 41 years old. # But also we can see an outlier, because the maximum value, 81 years old, is far from the previously mentioned values.

Q 10. Plot the distribution plot for the variable 'BloodPressure'. Write detailed observations from the plot. (2 Marks)

In [10]:

```
# remove _____ & write the appropriate library name

sns.displot(pima['BloodPressure'], kind='kde')
plt.show()
```

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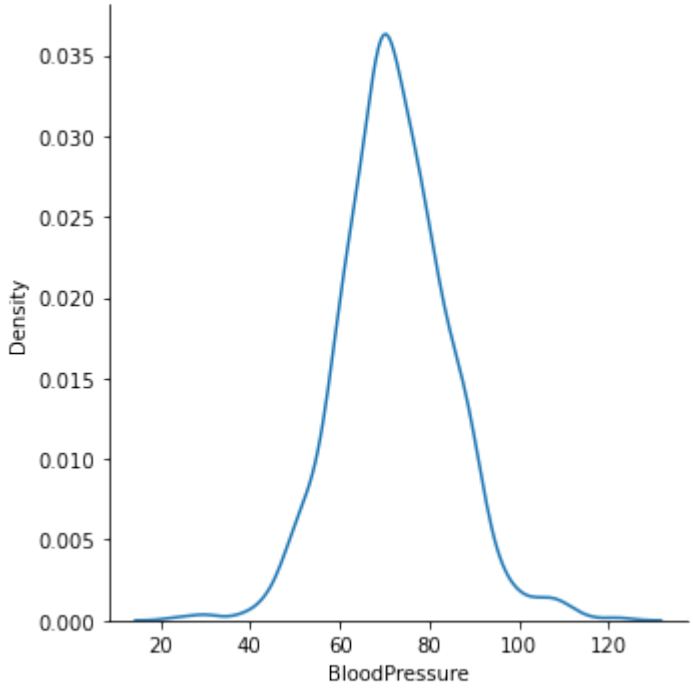

Write your Answer here:

Ans 10: sns.displot(pima['BloodPressure'], kind='kde') plt.show() # We can see that the mean value of blood pressure is about 70. # Most of the blood pressure values are between 40 and 100. # Very few people that participated have a blood pressure below 40 and above 100. # According to the medical literature, they have very low blood pressure. I think that some of them have hypotension. # According to the graph, I see that there are more people with blood pressure above 100 than people with blood pressure below 40.

Q 11. What is the 'BMI' of the person having the highest 'Glucose'? (1 Mark)

In [11]:
```
# remove _____ & write the appropriate function name

pima[pima['Glucose']==pima['Glucose'].max()]['BMI']
```

Out[11]:
```
661    42.9
Name: BMI, dtype: float64
```

Write your Answer here:

Ans 11: pima[pima['Glucose']==pima['Glucose'].max()]['BMI'] # The BMI of the person having the highest glucose is 42.9

Q12.

12.1 What is the mean of the variable 'BMI'?

12.2 What is the median of the variable 'BMI'?

12.3 What is the mode of the variable 'BMI'?

12.4 Are the three measures of central tendency equal?

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In [12]:
```
# remove _____ & write the appropriate function name

m1 = pima['BMI'].mean()  # mean
print(m1)
m2 = pima['BMI'].median()  # median
print(m2)
m3 = pima['BMI'].mode()[0]  # mode
print(m3)
```

```
32.45080515543617
32.0
32.0
```

Write your Answer here:

Ans 12: m1 = pima['BMI'].mean() # The mean is 32.45080515543617 print(m1) m2 = pima['BMI'].median() # The median is 32.0 print(m2) m3 = pima['BMI'].mode()[0] # The mode is 32.0 print(m3) # The three measures of central tendency are not equal, because the mean differs from the median and the mode. The median and the mode have the same value.

Q13. How many women's 'Glucose' levels are above the mean level of 'Glucose'? (1 Mark)

In [13]:
```
# remove _____ & write the appropriate function name

pima[pima['Glucose']>pima['Glucose'].mean()].shape[0]
```

Out[13]:
```
343
```

Write your Answer here:

Ans 13: pima[pima['Glucose']>pima['Glucose'].mean()].shape[0] # The answer is 343. 343 women's glucose levels are above the mean level.

Q14. How many women have their 'BloodPressure' equal to the median of 'BloodPressure' and their 'BMI' less than the median of 'BMI'? (2 Marks)

In [14]:
```
# remove _____ & write the appropriate column name

pima[(pima['BloodPressure']==pima['BloodPressure'].median()) & (pima['BMI']<pima['B
```

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Out[14]:

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction
14	5	166	72	19	175	25.8	0.587
93	4	134	72	20	79	23.8	0.277
103	1	81	72	18	40	26.6	0.283
205	5	111	72	28	79	23.9	0.407
299	8	112	72	20	79	23.6	0.840
325	1	157	72	21	168	25.6	0.123
330	8	118	72	19	79	23.1	1.476
366	6	124	72	20	79	27.6	0.368
380	1	107	72	30	82	30.8	0.821
393	4	116	72	12	87	22.1	0.463
406	4	115	72	20	79	28.9	0.376
446	1	100	72	12	70	25.3	0.658
460	9	120	72	22	56	20.8	0.733
488	4	99	72	17	79	25.6	0.294
497	2	81	72	15	76	30.1	0.547
510	12	84	72	31	79	29.7	0.297
568	4	154	72	29	126	31.3	0.338
615	3	106	72	20	79	25.8	0.207
635	13	104	72	20	79	31.2	0.465
644	3	103	72	30	152	27.6	0.730
717	10	94	72	18	79	23.1	0.595
765	5	121	72	23	112	26.2	0.245

Write your Answer here:

Ans 14: pima[(pima['BloodPressure']==pima['BloodPressure'].median()) & (pima['BMI']

Q15. Create a pairplot for the variables 'Glucose', 'SkinThickness', and 'DiabetesPedigreeFunction'. Write your observations from the plot. (4 Marks)

In [15]:

```
# remove _____ & write the appropriate function name

sns.pairplot(data=pima,vars=['Glucose', 'SkinThickness', 'DiabetesPedigreeFunction'
plt.show()
```

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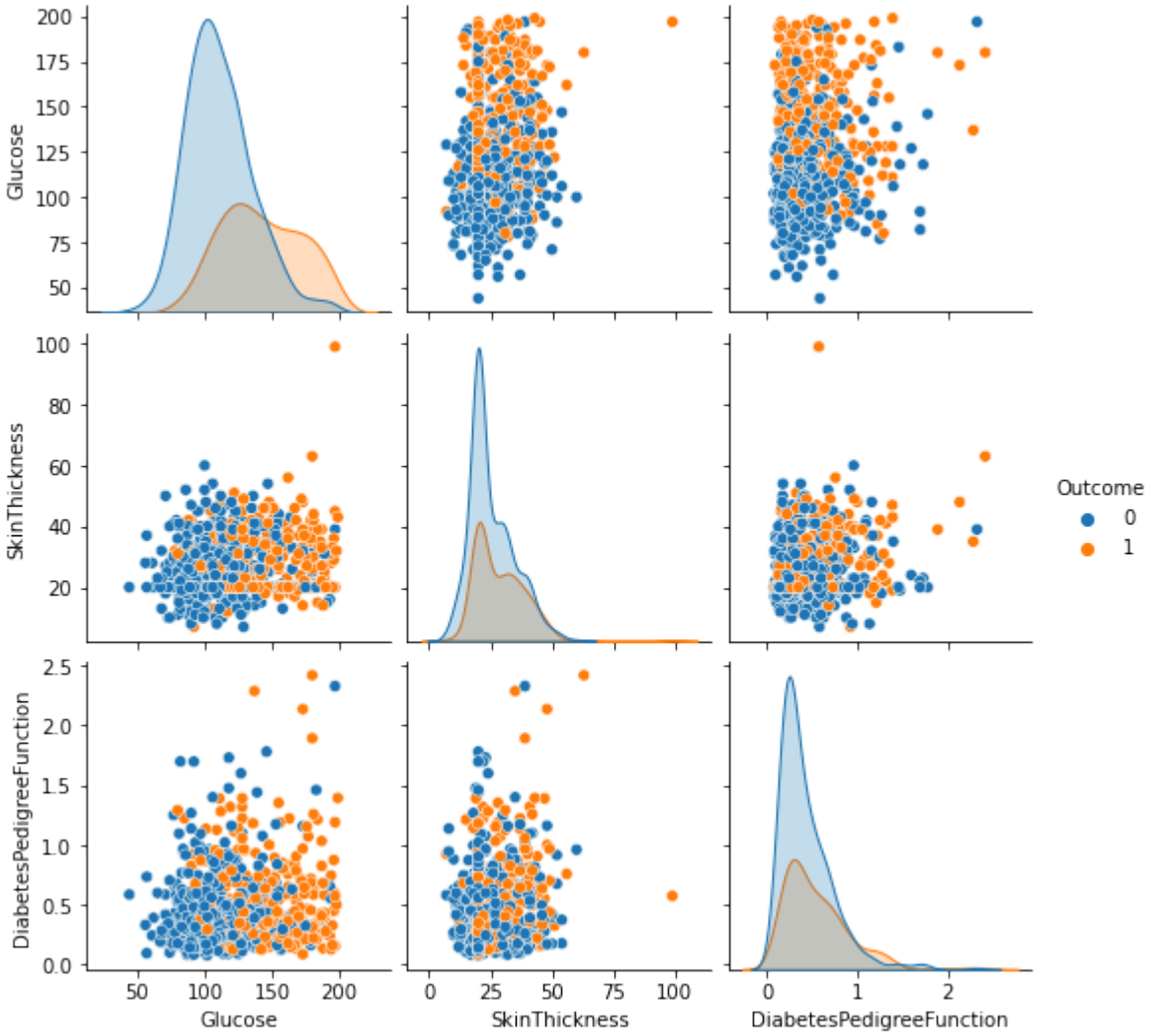

Write your Answer here:

Ans 15: sns.pairplot(data=pima,vars=['Glucose', 'SkinThickness', 'DiabetesPedigreeFunction'], hue='Outcome') plt.show() # Glucose graphs. Curiously, I found that there are more no diabetic women with high levels of glucose than diabetic women. For example, the no diabetic outcome reaches values near of 200, while the diabetic outcome reaches values near of 100. Maybe that's because people with diabetes try to lower their glucose values motivated by the fear of death. But also, it is interesting that there are a lot of people without diabetes with low glucose values (according to the scatter in the pairplot). # Skin thickness. No diabetic women - higher values of thickness than diabetic women. According to the medical literature, it is interesting that a lot of women without diagnosed diabetes have higher values of skin thickness. That's why it is important to see also the outliers in the scatter of the pairplot, because we can find some diabetic women with incredibly high values of skin thickness, maybe because their diabetes is worst. # Diabetes Pedigree Function. It is shockingly surprising that a lot of women without diabetes have a high likelihood of diabetes based on family history. Also, it is very interesting that in the two outcomes (with or without diabetes), there are outliers that reach the higher values of likelihood.

Q16. Plot the scatterplot between 'Glucose' and 'Insulin'. Write your observations from the plot. (2 Marks)

In [16]:
```
# remove _____ & write the appropriate function name

sns.scatterplot(x='Glucose',y='Insulin',data=pima)
plt.show()
```

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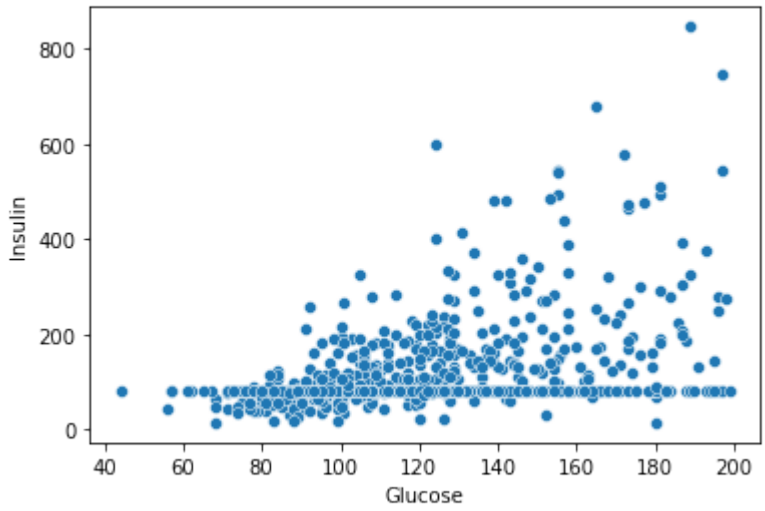


Write your Answer here:

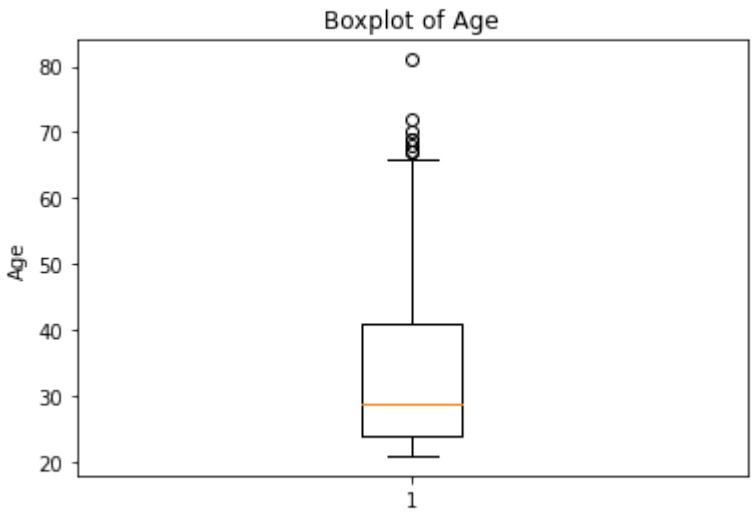
Ans 16: sns.scatterplot(x='Glucose',y='Insulin',data=pima) plt.show() # I found that higher the values of glucose, higher the values of insuline. # But, I have to recognize that there aren't a lot of outliers in the insuline values when glucose reach 160.

Q 17. Plot the boxplot for the 'Age' variable. Are there outliers? (2 Marks)

In [17]:
```
# remove _____ & write the appropriate function and column name

plt.boxplot(pima['Age'])

plt.title('Boxplot of Age')
plt.ylabel('Age')
plt.show()
```



Write your Answer here:

Ans 17: plt.boxplot(pima['Age']) plt.title('Boxplot of Age') plt.ylabel('Age') plt.show() # Yes, there are outliers.

Q18. Plot histograms for the 'Age' variable to understand the number of women in different age groups given whether they have diabetes or not.

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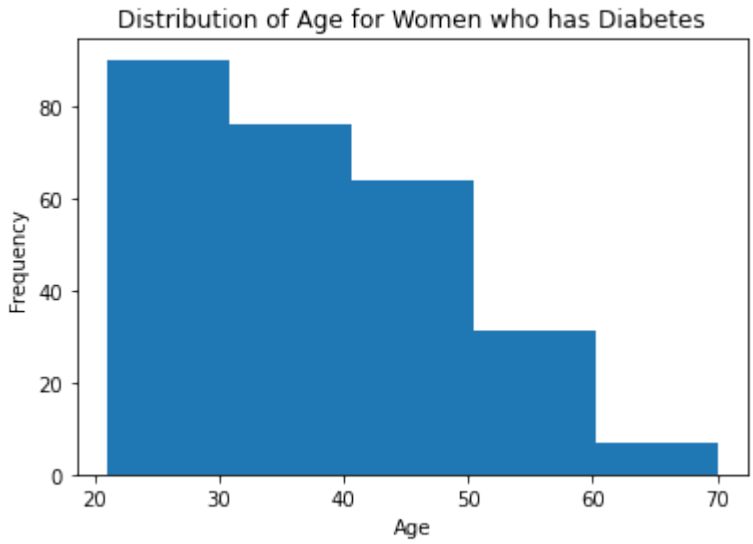
Explain both histograms and compare them. (3 Marks)

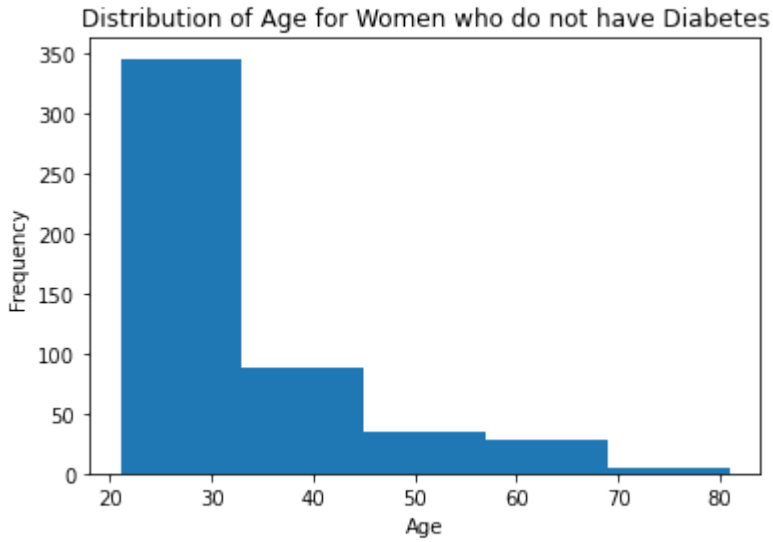
In [18]:
```
# remove _____ & write the appropriate function and column name

plt.hist(pima[pima['Outcome']==1]['Age'], bins = 5)
plt.title('Distribution of Age for Women who has Diabetes')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```



In [19]:
```
# remove _____ & write the appropriate function and column name

plt.hist(pima[pima['Outcome']==0]['Age'], bins = 5)
plt.title('Distribution of Age for Women who do not have Diabetes')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```



Write your Answer here:

Ans 18: plt.hist(pima[pima['Outcome']==1]['Age'], bins = 5) plt.title('Distribution of Age for Women who has Diabetes') plt.xlabel('Age') plt.ylabel('Frequency') plt.show() plt.hist(pima[pima['Outcome']==0]['Age'], bins = 5) plt.title('Distribution of Age for Women who do not have Diabetes') plt.xlabel('Age') plt.ylabel('Frequency')

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/omen who has Diabetes. A lot of women with diabetes have 20-50 years old. They are very young. Also, it is shockingly surprising that a lot of them have between 20 to 30 years old. #

Distribution of Age for Women who do not have Diabetes. A lot of women without diabetes have 20-30 years old. There are very few no diabetic women in the 30-80 range. # I think that there are more women without diabetes at the younger ages. But also, I think there are a lot of young women, because I found a lot with diabetes with 20-30 years old. Also, maybe it is very common to find women with 30-50 years old with diabetes.

Q 19. What is the Interquartile Range of all the variables? Why is this used? Which plot visualizes the same? (2 Marks)

In [20]:
```
# remove _____ & write the appropriate variable name

Q1 = pima.quantile(0.25)
Q3 = pima.quantile(0.75)
IQR = Q3 - Q1
print(IQR)
```

```
Pregnancies                  5.0000
Glucose                     40.5000
BloodPressure               16.0000
SkinThickness               12.0000
Insulin                     48.2500
BMI                          9.1000
DiabetesPedigreeFunction     0.3825
Age                         17.0000
Outcome                      1.0000
dtype: float64
```

Write your Answer here:

Ans 19: Q1 = pima.quantile(0.25) Q3 = pima.quantile(0.75) IQR = Q3 - Q1 print(IQR) # Pregnancies 5.0000 # Glucose 40.5000 # BloodPressure 16.0000 # SkinThickness 12.0000 # Insulin 48.2500 # BMI 9.1000 # DiabetesPedigreeFunction 0.3825 # Age 17.0000 # Outcome 1.0000 # The interquartile range is very useful to know the values between the first and third quartiles. Also, it is known that represents the middle 50% of data. # The box plot helps us to visualize the 25th and 75th percentile.

Q 20. Find and visualize the correlation matrix. Write your observations from the plot. (3 Marks)

In [21]:
```
# remove _____ & write the appropriate function name and run the code

corr_matrix = pima.iloc[:,0:8].corr()

corr_matrix
```

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Out[21]:

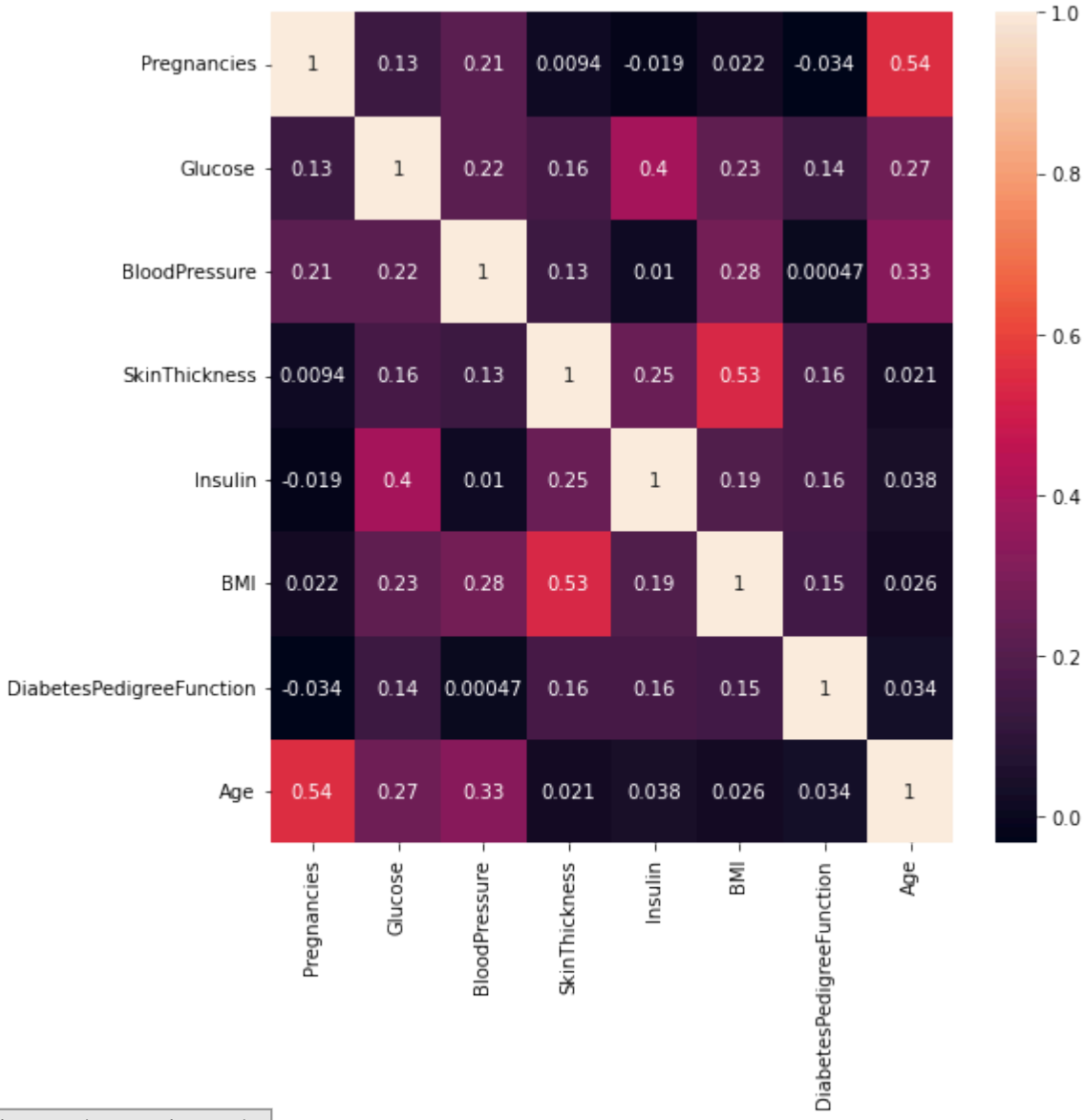
	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BM
Pregnancies	1.000000	0.128022	0.208987	0.009393	-0.018780	0.021546
Glucose	0.128022	1.000000	0.219765	0.158060	0.396137	0.231464
BloodPressure	0.208987	0.219765	1.000000	0.130403	0.010492	0.281222
SkinThickness	0.009393	0.158060	0.130403	1.000000	0.245410	0.532552
Insulin	-0.018780	0.396137	0.010492	0.245410	1.000000	0.189919
BMI	0.021546	0.231464	0.281222	0.532552	0.189919	1.000000
DiabetesPedigreeFunction	-0.033523	0.137158	0.000471	0.157196	0.158243	0.153508
Age	0.544341	0.266673	0.326791	0.020582	0.037676	0.025748

In [22]:

```
# remove _____ & write the appropriate function name

plt.figure(figsize=(8,8))
sns.heatmap(corr_matrix, annot = True)

# display the plot
plt.show()
```



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Write your Answer here:

Ans 20: # The heatmap shows the correlation between variables. # If the value is 1, the correlation is positive. There is a mild correlation between pregnancies and age (0.54). Also I found a mild correlation between the skin thickness and the BMI (0.53). There is an interesting relation between glucose and insulin (0.4), which is reasonable because the medical literature shows that insulin is a hormone made by the pancreas that helps glucose. # If the value is 0, there is no correlation. There is a little correlation between age and blood pressure (0.33). That's reasonable because some people with high blood pressure is older, but it is not a rule. There is a little correlation between glucose and blood pressure (0.22), the same with BMI and glucose (0.23). # There is very little correlation between diabetes pedigree function and blood pressure (0.00047), age and insuline (0.038), age and BMI (0.026), age and diabetes pedrigree function (0.034). # If the value is -1, the correlation is negative. I found a -0.019 between insulin and pregnancies.

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